NIL Investment Analysis Report

Plan

Business Objective

College Basketball programs are facing more financial pressire to allocate NIL resources effectively. Programs have to balance the financial resources available for player compensation with goal of getting a competitive roster from it. Understanding hpw NIL spending relates to projected team performance can help programs evaluate their financial position relative to other schools in their conference.

Our goal is to estimate team NIL as a function of BPR and conference affiliation. The model will generate an NIL benchmark for each team, allowing us to compare a program's estimated NIL with the level predicted for a team with similar characteristics

Schools with estimated NIL below their benchamrk may want to do further looks as pontentially cost-efficient programs, while those above the benchmark may want to do additional review of their resource allocation

Because the analysis relies on estimated NIL values and projected rather than realized basketball performance, these comparisons are descriptive rather than causal. A difference from the model benchmark does not, by itself, establish whether a program is overpaying or underpaying its players.

Decision Problem and Objective Function

Our decision problem is to establish a reasonable NIL amount for each Division I basketball program based on its team strength and conference affiliation. These benchmarks can then support the amount of teams whose estimated NIL levels differ susbstantially from comparable programs.

Let Y_i represent the estimated NIL spending of team i, measured in millions of dollars. Let X_i contain the team's BPR and conference. The action a_i is the NIL benchmark produced by the model for that team.

We ended up using quadratic loss:

$$\ell(a_i, Y_i) = (Y_i - a_i)^2$$

Quadratic Loss penalizes larger prediction errors more heavily then smaller errors, making it appropriate for constructing benchmarks where lots of money discrepancies are big consequences

Our objective is to have a function that minimizes that expected loss

$$f^* = \arg\min_f E[(Y - f(X))^2]$$

Under squared-error loss, the best prediction for a given set of characteristics is the mean:

$$f^*(X_i) = E[Y_i \mid X_i]$$

Because we dont know the true relationship, we estimate using regression:

$$\widehat{R}(f) = \frac{1}{n_{\text{test}}} \sum_{i \in \text{test}} (Y_i - \widehat{Y}_i)^2$$

After our model, we calcuate each team's estimated NIL gap:

$$G_i = \widehat{Y}_i - Y_i$$

A positive gap means that estimated NIL is below the model's benchmark, while a negative gap means estimated NIL is above the benchmark

These gaps are indicator for teams that should have further investigation rather definitive measures of financial efficiency. The available data do not establish the causal effect of NIL spending on basketball performance or determine an optimal roster budget. Furthermore, the observed outcome is a third-party NIL estimate rather than verified actual expenditure.

Here is an example:

```
# Quadratic loss for NIL predictions in millions of dollars
def nil_squared_loss(actual_nil, predicted_nil):
    return (actual_nil - predicted_nil) ** 2


# Illustrative prediction: $2.5M benchmark versus $3.0M estimate
nil_squared_loss(
```

```
    actual_nil = 3.0,
    predicted_nil = 2.5
)
```

```
0.25
```

Build

Ideal Data and Data-Generating Process

Model Specification

Simulating Data and Recovering Parameters

Explore

Data Sources and Collection

Data Cleaning and Preparation

Descriptive Statistics

Exploratory Data Analysis

Data Limitations

Reconcile

Validity and Representativeness

Linearity

Independence

Constant Variance

Normality

Identifiability

Model Refinements

Fit

Ordinary Least Squares

Frequentist Inference

Bayesian Inference

Evaluate

Model Fit and Interpretation

Model Comparison